

TELL, AND TWO ARGUMENTS ABOUT MOBILE

a one-thumb extraction run where the hunter learns you
faster than you learn the map

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PLAY <https://thegigglingpanda.github.io/tell/>
CODE <https://github.com/thegigglingpanda/tell>

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- Q1** **Build a playable prototype** — TELL: pitch, core loop, progression, money, AI, shipping, references, and the decision log
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- FIG** The loop, and eight runs of automated playtest
-
- Q2** **Gameplay insights & strategy** — why mobile extraction hasn't been cracked, and what a breakout would take
-
- Q3** **Design specification** — The Ledger, a feature for a mobile survivor-like
-
- WF** Three screens, annotated, and the interaction timing
-

The three answers are deliberately connected. Q2 makes a claim about why the genre fails on phones; Q1 is a playable attempt to test the smallest piece of it. Q3 is a separate problem, on purpose.

QUESTION 1 — TELL

Play it: <https://thegigglingpanda.github.io/tell/> — one click, nothing to install

Code: <https://github.com/thegigglingpanda/tell>

Immersive Hook: A one-thumb extraction run where the hunter learns you faster than you learn the map.

This prototype exists to test one claim I make in Question 2: that mobile extraction fails because the *tension* doesn't port, not because the shooting doesn't. So I built the tension without the shooting, to see whether it survives on its own.

Assumptions I made, since the brief left them open. Portrait, one hand, sessions under two minutes. No art budget — plain shapes, as allowed. No account and no server, which means the learning lives in the browser tab and dies when you close it. That last one is a real limitation and I'll be specific about what it means for the metagame section below.

THE PITCH

TELL is a top-down extraction run that lasts about ninety seconds.

You drop into the dark with a small circle of light. There are crates. Every crate you take is worth something, and every crate you take makes your light smaller. Four gates sit at the edges. Reach one and you bank the haul. Die and you drop everything you were holding.

There is one other thing down there. It cannot see in the dark either, so it learns you instead.

Who it's for. Players who already like the extraction feeling — the specific sickness of "one more crate" — and cannot give it ninety minutes and a mouse. On a mid-tier Android in India that describes a lot of people, though I'd want data before I put a number on it.

Why they'd play. Because something guessing right about you feels different from something chasing you. I don't have a better argument than that yet, which is why the prototype exists.

CORE LOOP AND FIRST SESSION

One question, asked roughly every twenty seconds: take one more, or leave?

Everything else is scaffolding for that question. Greed isn't abstract here. It's a vision penalty you watch shrinking around your own feet. Six crates in you are rich and nearly blind, and the gate is a long way through dark you can't read any more.

The first three minutes tend to go like this. Run one, you die confused. Run two, you work out that light and loot are the same currency spent in opposite directions. Run three, you extract with a small bag and feel clever about it. Run four, you get greedy.

What brings a player back is the card at the end of the run:

IT HAS LEARNED:

- You work the edges first. The middle frightens you.
- When I am close, you run in straight lines.
- You leave through the north gate. You always do.

Those lines are sentence templates. Which one fires, and whether it fires at all, comes from thresholds on counters of what you actually did — where you looted, which gate you banked through, how your heading changed when the Stalker got close. On run one, with a single data point, it says "*You left through the north gate. I will remember that,*" because claiming a habit from one observation would be a lie the player could catch.

Then it acts on what it has. When you're carrying a heavy load, it stops hunting and goes to wait at the gate you keep using.

So the day-1 hook and the core loop are the same object. I didn't attach a retention feature to a game — the thing that makes you press Again is the thing the game is about.

PROGRESSION AND METAGAME

The build contains one number and one meter: total banked, and how well it knows you. There is no persistence across browser sessions. Everything in this section beyond those two things is design I'd build next, not code you can play, and I'd rather say so than have you find out by refreshing the page.

The idea the build is testing: **in TELL, the progression belongs to the enemy.**

Most roguelites grow the player. Numbers rise, the run gets easier, the fantasy is mastery. TELL inverts that. You don't get stronger. Your opponent gets better informed, and the curve meant to carry months is the accumulation of a file on you.

What that would unlock:

- **The profile is the save file** — a behavioural fingerprint rather than a stash. In the shipped version it would persist to an account.
- **Counterplay is self-knowledge.** The only way to beat something that has read you is to stop being the person it read. Not aim, not routes: deliberately breaking your own habits. That skill never runs out of material.
- **Stalker variety as content.** One reads your routes, one reads your greed, one reads your panic. Same arena, different pressure.
- **The mirror.** A weekly run against a snapshot of your own profile from a month ago.

Two quantities have to stay separate here, and working that out was the most useful thing that happened during the build. *What it believes* about you should be able to change, sharpen, and go stale — a habit you've dropped ought to fade off the card. *How much it has figured out* should not fall, because a progression meter that goes down is a broken promise. The build keeps them apart. The three sentences shift between runs; the percentage only climbs, and it settles in the sixties rather than reaching a hundred.

The obvious failure mode is a difficulty spiral. If it always knows, you always lose, and people leave. The profile needs habit decay and the Stalker needs a confidence ceiling so it can be visibly wrong sometimes. It has to be readable rather than omniscient, and that's a tuning problem I haven't solved, only named.

MONEY

There's an obvious way to monetise this and I don't think it should be built.

The obvious way is to sell forgetting: a consumable that wipes one thing the Stalker knows about you. It would convert extremely well, because it's priced against relief. It would also be the worst decision available. The only asset this game has is that something knows you, and selling the erasure charges players to remove the reason they're there. Eighteen months of that and you have a maze game with a shop. I'd write the refusal into the design pillars so a bad quarter can't quietly reverse it, and I'll admit up front that the refusal costs real revenue.

Here's what I'd actually build, and I want to be straight that this is the weakest part of my answer.

The business is ads-first. At ₹-tier in India, on a game with ninety-second sessions and no PvP, rewarded video is where the money is and IAP is the tail. Pretending otherwise would be a nicer-sounding plan and a worse one. Two placements fit the loop:

- **Double your banked haul** after a successful extraction. Additive, opt-in, and it only appears when the player is already pleased.
- **A look at the arena layout** before you descend. Buys knowledge, which is the game's currency, and doesn't touch the Stalker's knowledge of you.

What I would not do is put an ad on death to recover the loot you dropped. That's the insurance trap again with a video attached.

Then IAP, in the order I'd expect revenue:

1. **Remove ads.** The highest-converting purchase in this market, and it sells comfort rather than advantage.
2. **Stalkers.** New readers with new reading styles, sold or granted seasonally. Horizontal content — it changes the pressure, never the player's power.
3. **A pass along the knowledge climb.** The meter rises whether you pay or not; the pass decorates it.
4. **Cosmetics on your light.** Lantern colour, falloff shape, the sound a crate makes. I'll be honest that this converts badly: it's a single-player game, so nobody else ever sees your light. It's on the list because it's cheap to produce, not because I expect it to earn.

The payer curve I'd expect is shallow and wide rather than whale-driven — no PvP means no status goods and no arms race, so there's no natural high end. That's a genuine constraint on the design and it's the thing I'd want to argue about, because it may mean this game is a ₹40-ARPPU product forever, and whether that's a business depends on install cost.

One rule underneath all of it: a player who has never spent must never lose a run to someone who has. Not as ethics. In India the top of the funnel is the business.

AI

How I used it to build this. All of the implementation. I worked in a long session with Claude: I set the constraints and made the design decisions, it wrote the JavaScript. The useful part was asking for counterpoints on every fork instead of agreement. Tap-to-move over a virtual stick came out of an argument I lost. Stating the evidence honestly on run one came out of one I won. Full log in the appendix, including four bugs its own automated playtest found in my design.

What I didn't outsource: what the game is about, what it feels like, what it refuses to sell, and what I'd kill.

Whether AI is part of the game. Yes, and this is the part I care about.

The Stalker builds a model of the player across runs — exit preference, looting territory, greed threshold, panic movement, whether they freeze when frightened — and acts on it before the player knows it exists. Then it states what it has learned in plain language.

I should be precise about what this is and isn't, because the honest description is still the interesting one. It's thresholded arithmetic over behavioural counters, not a model and not inference. A director that reads the player is old — Left 4 Dead shipped one in 2008. What I haven't seen done is the last step: **the system telling the player, in sentences, what it has concluded about them.** Directors have always worked invisibly, tuning pressure behind the curtain. Making the reading legible turns a difficulty system into a character, and it turns the player's own habits into the content.

That's also why it's cheap. No inference cost, no server, a few hundred lines of arithmetic and one sentence at the right moment.

SHIPPING

What I'd test first. One question: does the card make a new player press Again?

Twenty players, moderated, ten minutes each. I'd watch two things — whether their eyes go to the card or straight to the button, and runs per session. Twenty is enough to answer "*do they read it*" and nowhere near enough to answer "*does it retain*," and I'd resist letting the first stand in for the second.

What would make me kill it or turn it:

- Eyes go straight to Again and never to the card. Most likely outcome, and it's a legibility failure rather than a fun one, so it earns one redesign of the card before it earns a kill.
- They read it and shrug. Worse and mostly unfixable. I'd turn the project toward the greed-and-vision mechanic, which stands up without the AI.
- They read it and don't believe it. Tuning: fewer traits, higher confidence bars.

Two numbers at soft launch.

1. **Runs per first session.** Target six or more. Below four the loop isn't closing.
2. **D1 return, split by whether the player saw a correct trait in session one.** This is the actual experiment. To detect an eight-point difference against a ~30% baseline at 95% confidence I'd need roughly 550 players per cell, so about 1,100 installs routed into the test — which is why it's a soft-launch measurement and not something twenty people in a room can tell me.

I'd ignore ARPDau, session length and conversion for the first two weeks. There's no sense tuning an economy on a loop that hasn't proven it's fun.

Fun Factor: 6.8. Assigned, not measured. Held against five inputs:

INPUT	WHAT IT MEASURES	READING
Decision Density	meaningful choices per minute	High — a greed check every ~20s
Loss Weight	what a death costs	Medium — carried loot only
Legibility Delta	what you learn about it vs. what it learns about you	Negative, deliberately
Restart Friction	seconds from death to next run	~1s, one tap
Surprise Rate	novel events per run	Low, rising as the Stalker starts guessing right

The number's job is to be a fixed point that later playtests can move. Nothing here has been playtested outside my own hands and an automated harness, and I'd distrust any figure I claimed otherwise.

REFERENCE GAMES

Bullet Echo — the closest thing to this that shipped

I pulled this one apart hardest, because it had already answered the question I was most afraid of: whether darkness reads on a five-inch screen. It does. A light cone on a top-down map is instantly legible, and what you can't see becomes the content.

Borrowed: vision as the primary constraint instead of aim; short matches; top-down as the honest answer to small screens.

Done differently: in Bullet Echo, vision is a property of the hero you brought — chosen between matches, modified by gear and abilities, but not something you spend during a run. In TELL, vision is a resource you spend by being greedy. That turns an information constraint into a moral one. And Bullet Echo's threat is other players, so knowledge is roughly symmetric and dies with the match. TELL's threat is a system, and its knowledge accumulates across runs. That difference is the whole prototype.

BGMI and Free Fire — what mobile actually costs

A discipline borrow, not a mechanical one. These two taught me that on a mid-tier Android in an Indian summer, fidelity is a liability: the device throttles, the data costs money, and the session gets interrupted by an actual phone call. Free Fire's design is a long lesson in cutting fidelity and keeping the feeling — roughly fifty players and ten-minute matches where BGMI runs a hundred players across half an hour.

Borrowed: the assumption that the session survives interruption, one hand, and a hot device.

Done differently: I refused both shapes. Neither ten minutes nor thirty is the right container for the extraction feeling on a phone, so I compressed it to one antagonist and ninety seconds and kept only the greed.

Vampire Survivors and Survivor.io — restart friction, and how to lose it

Vampire Survivors' death-to-next-run gap is short and cheap, and that single property does more for retention than most metagames.

The instructive part is what happened on the way to mobile. Survivor.io took the same loop and put energy, a revive offer, a rewarded video and a results screen between you and the next run — turning a one-second gap into tens of seconds. Every one of those is a defensible monetisation decision, and together they tax the thing that made the loop work. That's the trade I was thinking about when I put TELL's restart at one tap and pushed the ads to the *end* of a good run rather than the start of the next one.

Done differently: both games make you a god by the last third of the run. TELL never lets you be one. Its power curve belongs to the enemy.

Baldur's Gate 3 — reactivity, and why mobile has to generate it

I keep going back to this one, and not to see the ending. I go back to test strategies, which is a strange reason to replay a story game and worth being precise about.

It isn't that the numbers get bigger on a second run. It's that the world answers differently. You solve the grove a different way and the game remembers, and forty hours later something small reflects it back at you. The feeling that produces is *this thing has been paying attention* — and it comes from the game **telling you it noticed**, through a companion's approval or a line of dialogue that calls back to a choice you'd half forgotten. Larian didn't hide the reactivity. They surfaced it constantly, in small pieces.

What I borrowed: the conviction that replay value can come from the system's response to you rather than from your own power growth, and that the response has to be *stated* to land.

What I did differently, and it's the whole reason TELL works the way it does: BG3's reactivity is authored. Thousands of hand-written branches, an enormous budget, and — crucially — finite. Play long enough and you've seen it. A ninety-second mobile run cannot author any of that. Nobody is writing ten thousand lines of reactivity for a session that fits between two bus stops.

So the only route left is to *generate* it from observation. Each individual piece is far shallower than a Larian callback — three plain sentences instead of a scene. But it's unbounded, and it's about the specific person holding the phone rather than about a character they picked. That trade — much less depth per instance, infinitely more of it, aimed at you personally — is the design bet TELL is making, and BG3 is where I got the idea that being noticed is worth building a game around.

XCOM — one lesson, stated small

XCOM shows you 85%, you miss, and you feel robbed — but you know exactly what happened and you go again. Hidden odds don't produce that. They produce confusion. It's why the Stalker states what it knows in sentences instead of letting you slowly wonder.

I've played XCOM, not dissected it. That's the whole of what I'd claim from it.

APPENDIX — AI USE, SOURCES, AND THE DECISION LOG

Tools. Claude wrote the implementation. Playwright ran automated playtests headlessly — a scripted bot playing consecutive runs with one deliberate habit, which is how four of the bugs below were found.

Division of labour. I set the constraints and made the design decisions, and asked for counterpoints on each fork rather than agreement.

FORK	CHOSEN	TURNED DOWN
Concept	Extraction micro-loop + adaptive stalker	A Ludo-derived skill loop; an AI-director-first arena
Controls	Tap-to-move, auto-loot	Virtual stick; auto-run with tap-to-stop
AI legibility	Explicit death card, plain language	Silent AI; delayed reveal after two runs
Monetisation	Refuse to sell forgetting; ads-first with remove-ads leading IAP	Selling forgetting as a rare consumable
Hosting	GitHub Pages off the code repo	itch.io

What changed, and why:

1. **The Stalker wasn't predictive until run three.** I was told plainly that a reader who plays twice would see a competent maze game — the biggest risk in the submission. It now models the current run's behaviour and intercepts from run one.
2. **The card was over-claiming.** With one data point it said "you always do." It now says "I will remember that."
3. **Extraction had no sound.** Found by the harness. The best moment in the game was silent.
4. **Behavioural sampling ran every frame**, so traits appeared within a fifth of a second of proximity and the ratios were noise. Now sampled at 250ms.
5. **The knowledge meter snapped to 100% by run six.** Made confidence-weighted.
6. **Then it went down** — 25% to 20% to 69% to 47% across a captured run, because it was reading whichever traits currently qualified rather than accumulated evidence. That bug forced the belief/knowledge split described above, which made the design better. It now reads 35, 51, 62, 64 and plateaus.

Sources. lilagames.com for studio and Project Black background; published founder profiles for studio history. Arena Breakout timeline in Question 2 from PCGamingWiki and MoreFun's own Steam announcement. No proprietary information, and I have not played Project Black.

Two things I'd want on the record. The performance claim in the README is a target, not a measurement — 62fps in headless Chromium on a desktop is not 60fps on a mid-tier phone, and I won't claim the second until it's been on one. And the JavaScript is not mine in the sense that I typed it. Every design opinion in this document is, and I'll defend all of them in a room.

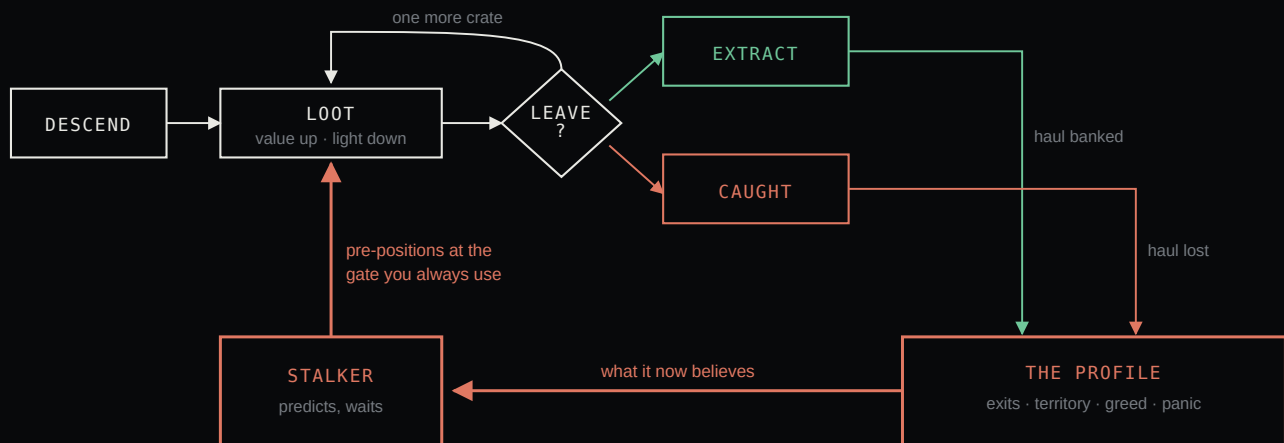
The loop, and the thing that closes it

Two figures. The first shows why this is a loop rather than a series of runs. The second is not a mockup — it is eight real runs of automated playtest, captured in order, with every line generated from what the bot actually did.

Figure 1 — one run resets. The profile doesn't.

Everything a player carries is lost at the end of a run. The only thing that crosses the boundary is the record of how they behaved, and it crosses into the enemy.

ONE RUN · ~90 SECONDS

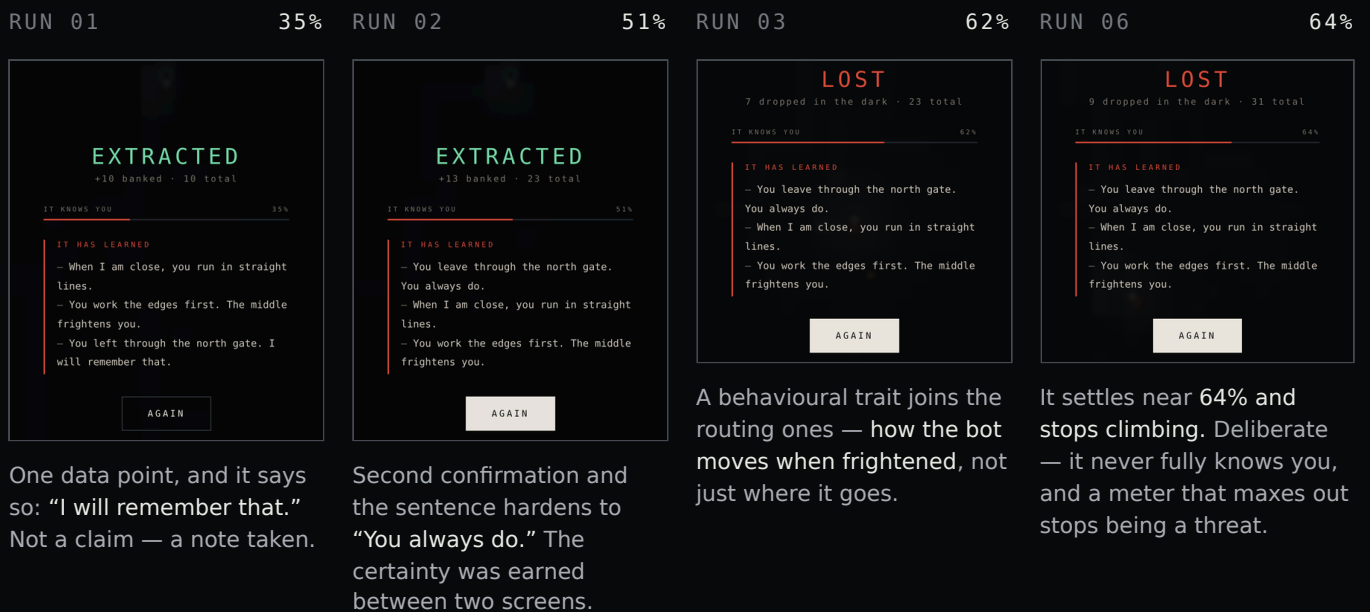


PERSISTS BETWEEN RUNS

The coloured path is the entire design. Take it away and this is a competent maze game with a greed mechanic. The player's loot, position and light all reset every ninety seconds; the record of their habits is the one thing that survives, and it survives on the enemy's side of the table. Progression belongs to the antagonist.

Figure 2 — eight runs, captured in order

An automated playtest with one deliberate habit: work the perimeter, always leave through the north gate. Nothing here is written by hand. Note the phrasing change between run 1 and run 2 — the system states how confident it actually is.



Fun Factor

6.8

ASSIGNED AT PROTOTYPE
NOT MEASURED

An imaginary anchor, held against five inputs so it can be argued with rather than asserted. Its job is to be a fixed point that later playtests can move — if the number can't drift, it isn't telling anyone anything.

INPUT	WHAT IT MEASURES	READING	WHY
Decision Density	meaningful choices per minute	High	a greed check roughly every 20 seconds
Loss Weight	what a death actually costs	Medium	carried loot only — the profile survives, so the run stings without the session hurting
Legibility Delta	what you learn about it vs. what it learns about you	Negative, by design	the differentiator; every other input exists to serve this one
Restart Friction	seconds from death to next run	~1s	one tap, no loading, no menu
Surprise Rate	novel events per run	Low, rising	flat until the Stalker starts guessing correctly — then it spikes

Figure 2 captured from an automated Playwright playtest against the submitted build, 8 consecutive runs, unedited.

Immersive Hook — a one-thumb extraction run where the hunter learns you faster than you learn the map.

QUESTION 2 — GAMEPLAY INSIGHTS & STRATEGY

THE GENRE: MOBILE EXTRACTION SHOOTERS

I'm arguing about a genre, not about anyone's game. I haven't played Project Black and I won't pretend to have opinions about a build I've never touched. Everything here is about the structural problem extraction has on phones, drawn from what has shipped publicly.

WHERE I'D START LOOKING

MoreFun Studios built *Arena Breakout* — the best-funded and best-executed mobile-native extraction shooter anyone has made. It launched in China in July 2022 and globally in July 2023, and technically it is a real achievement on phones.

By May 2024 they were running a closed PC beta. By August 2024 the PC version was in early access. It reached full release in China in April 2025 and Steam and Epic worldwide that September.

Roughly a year after the global mobile launch, the studio that had come closest to cracking mobile extraction was already building the PC version.

The obvious objection, which I'll make myself: Tencent ships cross-platform as a matter of course. PUBG, Delta Force and Naraka all exist on both. So platform expansion is the base rate, not a signal, and this is one data point read after the fact. I don't think it proves anything on its own.

I raise it because it's the most instructive *available* fact, and because it points at something I think is true for a reason that doesn't depend on it.

THE DIAGNOSIS: DREAD IS PRICED IN TIME, AND MOBILE CAN'T PAY

Here's what actually makes extraction feel like extraction.

Not the loot, and not the shooting. It's the moment deep into a Tarkov raid when you're carrying something good, the exit is a long way off, and you've already spent forty minutes you can't get back. That fear has a price, and the currency is invested time.

Now compress the session to three minutes, because that's what a phone is for. The loot is still there. The exit is still there. The fear isn't. Nobody is frightened of losing three minutes.

So studios substitute a different currency: gear value. If you can't make the player fear losing time, make them fear losing things. Bring your best rifle in, lose it forever.

That substitution is where I think the trouble starts, and it has three consequences.

Gear-priced dread needs a deep economy. For a rifle to be frightening to lose it has to be expensive, which means slow to earn, which means long grinding sessions — the exact thing mobile doesn't have. So the economy inflates and the sessions lengthen, and you find yourself building a PC game that happens to run on a phone.

It punishes newcomers hardest. A veteran loses a rifle they can replace in an hour. A new player loses everything they own, in their fourth match, to someone who has played four hundred. On PC, with a self-selecting hardcore audience, that cliff is the point. On mobile it empties the funnel before anyone converts.

And then the business model has to soften it. To stop the funnel draining you sell loss protection — gear insurance, buyback, safe-pocket slots, a second chance. Every shipped mobile extraction game I've looked at has some version of it, because the retention numbers demand it.

Look at what that purchase is, though. You spend the whole design manufacturing a fear of loss, and then sell the player a way not to feel it. The product and the purchase point in opposite directions, the players spending most are experiencing the game least, and the tension the design is built on becomes an inconvenience with a price tag.

That's the wall I think the genre keeps hitting. Not controls, not fidelity, not matchmaking.

The counter-argument I hear most is that mobile just needs better controls and the rest follows. *Bullet Echo* makes me doubt it: a top-down mobile shooter with vision constraints, tense and perfectly legible on a small screen, shipped years ago. Controls were never the ceiling. The pricing of fear was.

WHAT IT WOULD ACTUALLY TAKE

The move isn't to port extraction to mobile better. It's to stop importing the currency.

Mobile is poor in session time and gear depth. It's rich in two things PC isn't: frequency — dozens of tiny sessions a week — and identity, because the device knows exactly who you are every time you open it.

So price dread in knowledge instead of time.

1. Ninety seconds, as a constraint

Design the whole thing to fit between a lift arriving and a lift leaving. Systems that need longer get cut rather than compressed. Every other decision depends on this one, and most studios make it last.

2. The fragile asset is information, not inventory

In the prototype I built for Question 1, greed shrinks your field of vision — every valuable you pick up makes you blinder. You're not afraid of losing a rifle. You're afraid because you can't see, and you did that to yourself twenty seconds ago on purpose.

That dread costs nothing to manufacture. No economy, no grind, no inflation, and it arrives in run one.

3. The persistent object is a profile

This is the part I'd bet the project on. Instead of accumulating gear between runs, the game accumulates a model of the player — where they loot, which exit they trust, how greedy they get, what they do when frightened. The model belongs to the antagonist and makes it better at hunting that specific person.

Three things follow. Progression that never inflates, because the player doesn't get stronger and there's no power curve to balance. Content that can't be exhausted or wikied, because the content is the player's own behaviour. And a retention hook that *is* the core loop — the game tells you, in a sentence, what it has worked out about you.

The thing I'd have to solve is the spiral. If it always knows, you always lose, and people leave. It needs habit decay and a confidence ceiling so it can be wrong sometimes.

4. Social, but asynchronous — and not for the reason people assume

Extraction's best stories come from other players, and I want to be careful here, because the lazy version of this argument is wrong. India runs some of the largest live mobile PvP audiences on earth. BGMI and Free Fire prove the networking works at enormous scale.

What they don't do is ninety-second sessions. Matchmaking, lobby and load overhead is fixed cost, and at a ten- to thirty-minute session you can absorb it. At ninety seconds it eats the session. So the constraint is session length, not connectivity.

Which means: leave traces instead. Where other players died today. Which exit the server favoured this week. The last three sentences the antagonist learned about somebody else. It's cheap, it tolerates bad networks, and it still produces the sense that someone was here. I'll be honest that no mobile shooter has carried its social loop on this alone, so it's the least proven idea in this essay.

5. Don't sell relief from the core tension

A line rather than a preference. The moment you sell forgetting — a consumable that wipes what the antagonist knows about you — you've rebuilt the insurance trap with new art. Sell comfort instead: removing ads, new antagonists, cosmetics. Write it into the pillars so a bad quarter can't reverse it, and accept that the refusal costs revenue.

6. Build for the phone that exists in this market

Mid-tier Android, hot, on shared data. Sixty frames on plain shapes beats thirty on beautiful ones. Entry price points under ₹100. And a player who has never spent should never lose a run to someone who has — in India the top of the funnel is the business.

THE NUMBERS I'D HOLD IT TO

For the first month I'd refuse to look at revenue and watch three things.

1. **Runs per first session.** Target six or more. The only real test of whether ninety seconds is fun; below four the loop isn't closing and nothing downstream saves it.
 2. **D1 return, split by whether the antagonist read the player correctly in session one.** This is the experiment. Detecting an eight-point difference against a ~30% baseline needs roughly 550 players per cell, so it's a soft-launch measurement rather than a playtest one. If both cohorts return at the same rate, the thesis is wrong and I want to know in week one.
 3. **Share of revenue from comfort versus advantage.** If advantage purchases climb past a fifth, the insurance trap is growing back and the design is losing to the P&L.
-

WHAT I'D EXPECT TO BE WRONG ABOUT

Gear. Loot lust is very old and very reliable, and I'm proposing to run the genre without it. It's possible that wanting the shiny thing beats being known, for most players, and that the knowledge idea is a designer's pleasure rather than a player's. Ninety seconds is the cheapest way I know to find out.

The revenue ceiling. No PvP means no status goods, no arms race, and no natural high end to the payer curve. That might make this a shallow-and-wide business forever, which is a real problem and not one I can design my way out of.

Sources

- Arena Breakout mobile release dates (China July 2022, global July 2023): Wikipedia.
- Arena Breakout: Infinite PC timeline (closed beta May 2024, early access 13 August 2024, China full release 29 April 2025, Steam/Epic worldwide 15 September 2025): PCGamingWiki and MoreFun's own Steam announcement.
- Bullet Echo (ZeptoLab), BGMI, Free Fire, Escape from Tarkov: referenced from play, not from data.
- No revenue estimates are used here. Third-party app-market figures for Arena Breakout appear to exclude its Chinese storefront, where the game launched first, so I don't trust them enough to build an argument on.

QUESTION 3 — DESIGN SPECIFICATION

THE LEDGER

A single feature for a mobile survivor-like

One line: every permanent upgrade you take makes you stronger *and* feeds the horde in a dimension you chose, so you author your own late game.

Assumptions, since the brief left them open: fifteen-minute maximum run, portrait, one thumb, auto-attacking combat, a level-up card screen every 40-60 seconds — so roughly twenty picks across a full run. Mid-tier Android target. If those are wrong the feature survives; the tuning wouldn't.

1. WHY THIS FEATURE

The last third of a survivor-like run is the part players stop looking at.

The arc is consistent across the genre: a nervous opening, a thrilling middle as the build comes online, then several minutes of holding a phone while a screenful of things dies without threatening you. The run doesn't end because you were beaten. It ends because the clock ran out. Watch anyone play and you'll see them glance away, check a message, put the phone face-down and pick it back up.

There's a second problem that's quieter. Within a week of launch there's a best build, it's on YouTube, and the card screen turns into a hunt for three cards you already know you want. The genre's central mechanic — a real decision every minute — decays into pattern-matching.

Both come from the same place: **power is free**. You accumulate it, nothing pushes back, and the only thing between you and invincibility is a fixed spawn curve.

The Ledger makes power cost pressure, and lets the player choose which kind. That changes the card screen from "which buff is biggest" into "what fight do I want to be in at minute twelve." Same number of taps, same session length, different decision underneath.

2. THE SYSTEM

2.1 The core rule

Every upgrade card carries two lines: **GIVES** and **FEEDS**.

IF THE UPGRADE GIVES YOU...

IT FEEDS...

THE HORDE GAINS

Power — damage, crit, area

BULK

enemies with more health

Speed — fire rate, projectiles, movement

SWARM

more enemies, spawning faster

Life — max HP, armour, regen, lifesteal

FURY

enemies that move faster and chase harder

Each pick adds points to exactly one axis. Three axes, one sentence, no exceptions.

The Ledger prices permanent power only. One-shot consumables, pickups and heals feed nothing. That distinction matters for the fourth option below.

2.2 The fourth option: **DECLINE**

Every card screen offers three cards and a fourth choice. **DECLINE** takes no upgrade, feeds no axis, and restores a small amount of health.

This is the rule that makes the rest of the feature work, and it's the one I'd fight to keep if the scope got cut. Without it the level-up is mandatory, the player has no way to refuse pressure, and every claim below about self-balancing is false.

With it, the question at every screen becomes: *is this upgrade worth the fight it builds?* Sometimes it plainly is. Sometimes you're at a third health with SWARM already high and the right answer is to take the heal and stay where you are.

2.3 The Ledger readout

Three short bars in the HUD, top-right. BULK, SWARM, FURY. They only ever rise.

They aren't a threat meter. They describe the fight the player has built, and players learn to read them the way you read a board — not "am I losing" but "what shape is this."

2.4 Scaling, and why there is no cap

The obvious way to keep this safe is a hard cap per axis. That produces a cliff: with twenty picks available, an aggressive build saturates the cap around pick eight, after which power arrives free again and the feature stops working at exactly the minute it exists to fix.

So the escalation is asymptotic instead:

```
effect(n) = ceiling × (1 - 0.82n)
```

Each point adds about 82% of what the previous one added. Twenty points into one axis lands around 98% of that axis's ceiling without ever reaching it, and pick twenty still adds something. Power never becomes free again, and it never becomes impossible either.

FURY gets the lowest ceiling of the three. Enemy speed is the least forgiving variable on a touchscreen — you can't out-thumb it the way you can out-position it with a mouse.

2.5 Surplus, and how it stays honest

A system that only punishes commitment teaches hedging, and hedging is dull. So commitment pays.

When one axis sits **four or more points above your lowest**, enemies of that axis start dropping **Surplus**. Ten Surplus buys a **Windfall** card: full effect, and its FEEDS line reads **YOUR WEAKEST** — it adds one point to your lowest axis.

That last detail is doing specific work. My first version of this made Windfall feed nothing at all, which produced one dominant line of play: dump an axis, ride out the pain, then convert it into unlimited free power. Because Windfall raises your lowest axis, cashing in narrows your spread — and narrowing your spread switches off your own Surplus income. The strategy is self-limiting rather than optimal.

Two Windfalls per run maximum, as a belt on top of that.

2.6 Where the money touches it

Survivor-likes monetise through permanent meta-progression, so the Ledger has to answer for that or it quietly dies at the top of the payer curve.

Meta upgrades feed the Ledger at half rate. A player who has bought permanent power starts each run stronger, and their horde starts correspondingly heavier. The system stays intact, spending is still clearly worth it, and the Ledger can't be bypassed with money.

Half rate rather than full because a paying player should feel the benefit, and rather than zero because a paying player who is exempt from the game's central mechanic is playing a different, worse game.

The KPI hypothesis this feature carries is narrow and testable: *runs completed to the fifteen-minute mark should fall, and sessions per day should rise*. If people are dying more and playing more, the late game has stopped being dead air. If completion falls and sessions fall with it, the pressure is mistuned and the ceilings come down.

Telemetry I'd want from day one: axis distribution at run end, DECLINE rate by level-up index, time-to-death by Ledger shape, Windfall conversions per run, and the distribution of final spread. That last one tells me whether players are committing or hedging, which is the single thing the whole design turns on.

3. THE UPGRADE CHOICE MOMENT, SECOND BY SECOND

This is the fully-detailed section. It's the moment the feature lives or dies in, it happens about twenty times a run, and it has roughly two seconds of the player's attention each time.

3.1 What has to be true

The player has just levelled up mid-fight. Their thumb is somewhere over the lower third of the screen. Enemies are still on screen behind the card layer. They're mildly stressed, they may be standing on a bus, and they've decided — without knowing they decided — to spend no more than two seconds here.

In those two seconds they need three things: what each card gives, what each card feeds, and which of the three bars is about to grow. If any of it needs a second look, the feature is dead weight and the player starts tapping the leftmost card forever.

3.2 The sequence

0ms — the freeze. Combat pauses. The playfield desaturates to about 30% but is not blurred and not covered. The player has to still see the horde they're about to make worse. Deciding while looking at forty enemies produces a different decision than deciding on a blank screen, and the first is the one we want.

0-120ms — the Ledger rises. Before any card appears, the three bars scale to roughly 1.6× and move from the corner toward the upper centre. The bars arrive first on purpose: the question gets asked before the options are offered.

120-260ms — cards enter. Three cards rise from the bottom edge, staggered 40ms apart, settling into the lower two-thirds. They stop short of the bottom 12% of the screen, which stays clear for the resting thumb — nothing tappable ever sits under a finger that's already down. DECLINE sits directly beneath the three cards and directly above that clear band: a low, wide strip, easily reachable, deliberately less prominent than the cards.

On settle — the projection. Each card draws a faint ghost segment onto its corresponding Ledger bar: this is how much that bar grows if you take me. Three cards, three ghosts, three different bars, all visible at once. The player is looking at a picture of three possible futures and doesn't have to read a number to compare them.

On touch-down — commitment preview. The moment a finger lands, that card's ghost goes solid and the other two vanish. The chosen future resolves. This is also the safety mechanism: the action fires on release, not on press, so a mis-touch can be cancelled by sliding off the card. Mis-taps in this genre are constant, and a mis-tap here is unrecoverable, because the bars never fall.

Touching DECLINE dims all three ghosts instead of solidifying one — the visual statement that nothing is going to grow.

On release — the resolve. The card flashes at its axis colour, the bar grows over 180ms with a short overshoot, the bars retreat to HUD size, combat resumes. About 340ms from release to control. Twenty of these per run; any longer and it starts to feel like paperwork.

3.3 Card anatomy

Reading order top to bottom, because a thumb-held phone is read top-down and the give is what the player came for:

1. **Name.** Two or three words. "Twin Shot." "Heavy Rounds."
2. **GIVES** — the effect in plain player language. "Two projectiles instead of one," never "+1 projectile count."
3. **A hairline rule.** The hinge between reward and cost. Cheapest and most important pixel on the card.
4. **FEEDS** — the axis name, its colour, and its magnitude as pips rather than digits. **SWARM** ■■ parses faster than "+2 Swarm," and pips stay legible at arm's length on a scratched screen in daylight, which is where this game gets played.

The axis colour is the load-bearing element. It ties the card to its bar to the enemies that arrive because of it, and once a player has connected those three they stop reading the FEEDS line and just recognise the colour. Read with your eyes in run one, with your reflexes by run five.

3.4 Teaching it without a tutorial

Nobody reads tutorials on a phone, so the first run teaches by constraint.

- **Level-ups 1, 2 and 3 offer single-axis cards only** — three Power cards, then three Speed, then three Life. The same bar grows all three times and the player cannot help noticing. Causality established with no text.
- **The first time an axis crosses three points**, the change is announced in four words at the top of the screen, in that axis's colour: "*THEY ARE HEAVIER NOW.*" Once per axis per run.
- **DECLINE appears from level-up 2**, not level-up 1. The player should experience taking something before they're offered the chance to refuse.
- **Level-up 4 is the first mixed offer.** By then the rule is in place and the first real decision means something.

3.5 Edge cases

CASE	BEHAVIOUR
Two level-ups within 2s	Queue. Never stack two card layers — resolve fully, then present the next.
Player idles on the card screen	No timer. The bus arrives, the phone comes down. Punishing the pause punishes the platform.
All three cards feed the same axis	Allowed and deliberate, roughly one offer in eight. With DECLINE present it's a real fork rather than a forced tax.
Player declines repeatedly	Permitted. They'll face a soft horde with a weak build and lose to the baseline spawn curve. That's the honest floor of the system, and it's where cautious players naturally settle.
Windfall available	Replaces the centre card, visually distinct, FEEDS reads YOUR WEAKEST .
Interruption — call, notification	Card state persists. Return to the frozen screen, not the fight.

3.6 Accessibility

Colour carries a lot here, so it can't work alone. Each axis also has a fixed **icon** and a fixed **bar position** — BULK left, SWARM centre, FURY right, always. A player who can't distinguish the three colours uses position and shape and loses nothing. Pips give a third redundant channel. Text never below 13sp, nothing tappable under 48dp.

4. WHAT THE PLAYER TAKES AWAY

The run summary shows the horde you built: three final bars and one sentence naming the fight. *"You chose to be fast, and to be swarmed."*

Two players who ran the same character for the same fifteen minutes get two different sentences. Comparing them is a conversation, which is the argument for doing this instead of putting a difficulty selector on the front screen.

5. RISKS A PLAYTEST WOULD HAVE TO SETTLE

Cognitive load. Three bars, two lines per card, a fourth option and a live fight is a lot for a two-second decision on a small screen. If testers can't state the rule after five minutes, I'd cut FURY and ship two axes. Two is worse and it still works, and I'd rather ship the smaller version than defend the elegant one.

Framing. Some players will read FEEDS as the game taking something from them, and feel meanly treated. The copy has to sell authorship rather than tax. The run-summary sentence is where that's won, which is why I'd write it before writing the tuning table.

And the one I actually don't know. I'm proposing that players enjoy choosing their own suffering. Roguelike audiences do it happily; a mass-market mobile audience with a five-minute attention span is a different animal and I don't have evidence either way. Two hundred players and a week would tell us, and I'd want that before a single number in §2.4 got tuned.

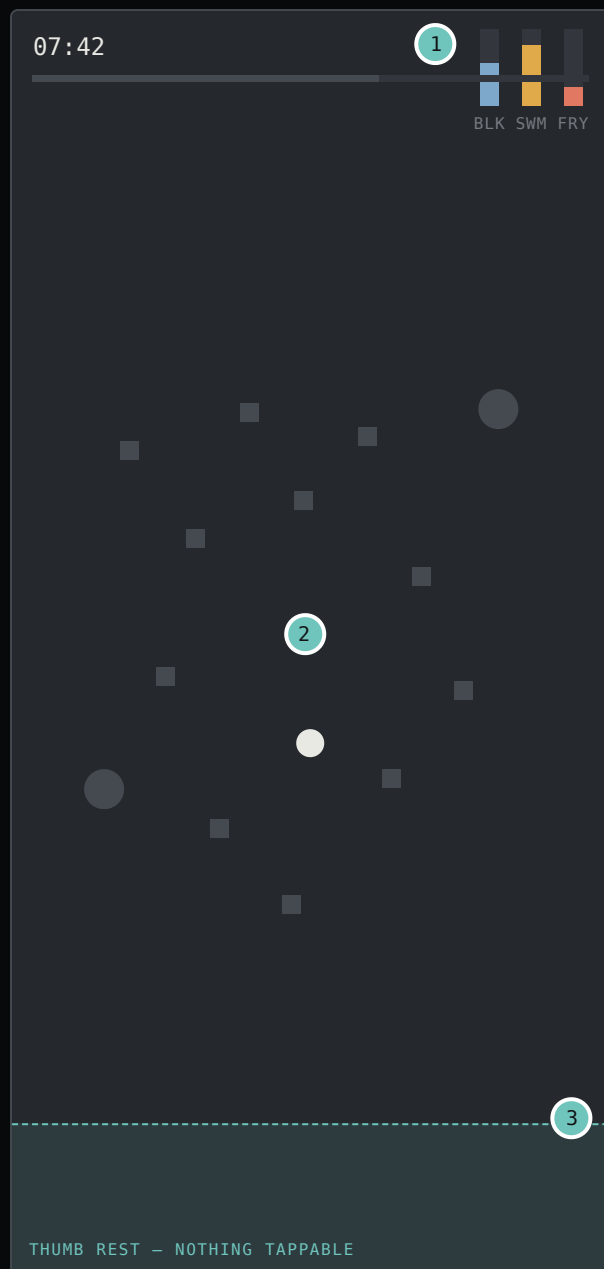
Wireframes follow: (1) in-run HUD with the Ledger, (2) the upgrade choice screen — the moment detailed in §3, (3) run summary.

The Ledger

Every upgrade makes you stronger and feeds the horde in a dimension you chose. Three screens: where the Ledger lives, the two seconds where the decision happens, and what the player carries out of the run.

■ Greybox – placeholder art, no visual design intended ■ BULK · enemy health ■ SWARM · enemy count
■ FURY · enemy speed

01 — IN-RUN HUD, MINUTE 7



- 1 The Ledger sits in the HUD, always visible. Three bars, never falling. It is not a threat meter — it describes the shape of the fight the player has built.
- 2 The horde reflects the bars. This run is SWARM-heavy and FURY-light: many small enemies, slow. A BULK run at the same minute would show six large ones instead.
- 3 Bottom 12% stays clear. Reserved for the resting thumb so the card screen never places a target under a finger that is already down.



- 4 The ghost segment is the whole feature. The finger is down on *Second Skin*, so FURY shows a translucent preview of the growth it would cause. Before touch, all three cards show their ghost at once — three futures, comparable without reading a number.
- 5 The playfield stays visible, dimmed to ~42%. Not blurred, not covered. Deciding while looking at the horde you are about to worsen produces a better decision than deciding on a blank screen.
- 6 Card reads top-down: name, GIVES, hairline, FEEDS. Cost is expressed as coloured pips, not digits — faster to parse, legible in daylight on a cracked screen.
- 7 Fires on release, not on press. A mis-touch can be cancelled by sliding off the card. The bars never fall, so an accidental commit is unrecoverable — this is the one safety mechanism the feature genuinely needs.
- 8 DECLINE is the rule that makes the rest work. Take nothing, feed nothing, recover a little health. Without it the level-up is mandatory, the player can never refuse pressure, and the self-balancing claim in §2.2 is false. Touching it dims all three ghosts rather than solidifying one.

THE LEDGER CLOSED

You chose to be fast, and
to be swarmed.

BULK		5
SWARM		8
FURY		2

SURVIVED 13:20 · 4,182 KILLED
SURPLUS SPENT 2

- 9 One sentence, generated from the final bars. Two players who ran the same character for the same fifteen minutes get two different sentences. That difference is a conversation, which a difficulty slider can never be.
- 10 Surplus spent is reported, not celebrated. Windfall cards cost Surplus and raise your weakest axis by one, so cashing in narrows your spread and throttles your own income. Self-limiting rather than optimal.

The two seconds, timed

Screen 02 in sequence. Total from release to player control: ~340ms. Roughly twenty of these per run.

TIME	WHAT HAPPENS	WHY
0ms	Freeze. Playfield desaturates to 42%.	The horde stays legible. The cost is decided in front of the thing it affects.
0–120ms	Ledger rises to 1.6× and moves toward upper-centre.	The bars arrive before the cards. The question is asked before the options are offered — this does the teaching a tutorial would.
120–260ms	Three cards enter from the bottom, staggered 40ms.	Staggering gives reading order for free without an animation the player has to wait out.
on settle	Ghost segments appear on all three bars simultaneously.	Three possible futures, side by side, compared visually rather than arithmetically.
touch-down	Chosen ghost solidifies; other two vanish.	Confirms intent before commitment. Slide off to cancel.
release	Flash, bar grows 180ms with overshoot, Ledger retreats, combat resumes.	About twenty of these across a fifteen-minute run. Any longer and it reads as paperwork.

Greybox wireframes · no art direction implied · portrait 9:19, mid-tier Android target